

START

```
graph TD; Start([START]) --> Init[INITIALIZATION<br/>GPIO, I2C, ADC, SPI, RTC,<br/>CONSTANTS]; Init --> ReadAcc[READ ACCELERATION<br/>VIBRATION<br/>I2C SAMPLE; ADXL345]; ReadAcc --> PostProc[VIBRATION ACCELERATION<br/>POST-PROCESSING<br/>VELOCITY, RMS VELOCITY, FFT]; PostProc --> ReadTemp[READ TEMPERATURE<br/>DIGITAL SAMPLE]; ReadTemp --> ReadCur[READ CURRENT<br/>ADC SAMPLE; ACS712]; ReadCur --> CondSig[CURRENT SIGNAL<br/>CONDITIONING<br/>BPF, SIGNAL AMP]; CondSig --> ReadHall[READ HALL EFFECT<br/>ADC SAMPLE; A3141+MAGNET]; ReadHall --> A((A))
```

INITIALIZATION
GPIO, I2C, ADC, SPI, RTC,
CONSTANTS

READ ACCELERATION
VIBRATION
I2C SAMPLE; ADXL345

VIBRATION ACCELERATION
POST-PROCESSING
VELOCITY, RMS VELOCITY, FFT

READ TEMPERATURE
DIGITAL SAMPLE

READ CURRENT
ADC SAMPLE; ACS712

CURRENT SIGNAL
CONDITIONING
BPF, SIGNAL AMP

READ HALL EFFECT
ADC SAMPLE; A3141+MAGNET

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